SCHEMATIC LOCUS DETECTED ON STRICT HENSELIAN LOCAL FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4097-4108, following the statement at 4089-4095.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Limits that are schemes. lemma-limit-is-scheme (same chapter, cited at ~4105-4107). Applied to X x_S Spec(O_{S,s}) = lim X x_S U.

Source reference. Original lemma 07SR: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1435–1435. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a quasi-compact and quasi-separated morphism from an algebraic space to a scheme S. If for every $x \in |X|$ with image $s = f(x) \in S$ the algebraic space $X \times_S \mathrm{Spec}(\mathcal{O}_{S,s})$ is a scheme, then X is a scheme.*

Proof. Let $x \in |X|$. It suffices to find an open neighbourhood U of $s = f(x)$ such that $X \times_S U$ is a scheme. As $X \times_S \mathrm{Spec}(\mathcal{O}_{S,s})$ is a scheme, then, since $\mathcal{O}_{S,s} = \mathrm{colim}\, \mathcal{O}_S(U)$ where the colimit is over affine open neighbourhoods of s in S we see that

$$X \times_S \mathrm{Spec}(\mathcal{O}_{S,s}) = \lim X \times_S U$$

By Lemma 07SR we see that $X \times_S U$ is a scheme for some U. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.